INDIVIDUAL REPORT

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INTRODUCTION

Vodafone is a prominent global telecommunications corporation, operating across Europe, Africa, the Middle East, Asia Pacific, and the Americas. Broadband internet access, television services, and mobile and fixed-line phone, data, and messaging services are just a few of the many services offered by the corporation.

With a solid track record of profitability, Vodafone is a financially stable business. Vodafone brought in €45.7 billion in revenue and €14.7 billion in adjusted EBITDA during the most recent fiscal year. Investor optimism in the company's future prospects is reflected in the recent increasing trend in the share price of the company.

Vodafone has experienced a lot of significant developments in the last few years, such as the selling of its ownership in the international tower infrastructure business Vantage Towers and the purchase of Liberty Global's Kabel Deutschland division. These modifications are a result of a larger strategic initiative to streamline the organisation and concentrate on the company's primary markets.

	Value
Revenue	€45.7 billion
Adjusted EBI	€14.7 billion
Net profit	€12.3 billion
Share price	Metric

All things considered; Vodafone is a profitable telecoms firm in a solid position. Despite a number of obstacles, such as growing competition from new competitors and growing expenses, the company is well-capitalized and has a capable management team in place (Anon., 2023).

DYNAMICS OF INVESTMENT

There are five distinct investment options, each with a projected cash flow of five years. The risk of each of the five investment opportunities is the same. Considering that we need to invest the 45 billion dollars in revenue from last year in one or more projects. The following is the initial investment and ranking for each of the five projects.

	Initial Investment	Profitability Index	Rank	NPV
Project A	-525000	£1.49	1	£255,896.50
Project B	-375000	£1.22	4	£84,223.18
Project C	-448000	£1.15	5	£66,194.85
Project D	-625000	£1.40	2	£249,737.63
Project E	-254000	£1.35	3	£89,722.89

The suggested project, from which the company may benefit more, is shown in the table along with its NPV and profitability index ranking. The business can invest in the project with the highest net present value (NPV) or in numerous projects at once based on the cash flow of each project. Every endeavour has the ability to succeed.

INVESTMENT APPRAISAL

The study includes the following investment appraisal techniques: NPV (Net Present Value), IRR (Internal Rate of Return), PP (Payback Period), ARR (Accounting Rate of Return), &WACC (Weighted Average Cost of Capital).

NPV is a flexible metric that may be applied to a variety of projects and investments. Although it is a very easy calculation to do, the results might give important information about how profitable a project or investment will be. Because it is a robust measure of profitability, variations in the discount rate have no effect on it.

It is predicated on several assumptions, such as the precision of the cash flow estimate and the discount rate, as well as variables like risk and uncertainty that influence investment decisions.

The discount rate known as IRR causes a project's net present value to be equal to zero. The iterative calculation of the IRR rate involves adjusting the discount rate until the net present value (NPV) is equal to zero. ARR is a straightforward indicator of a project's profitability that is computed by dividing the average yearly profit of an investment by the amount of the initial investment. Projects that generate cash flows early in their lifetimes and projects that generate cash flows later in their lives are not distinguished by ARR. The most advanced method for determining an investment's profitability is the probability measure. A favourable return on investment is indicated by a PI greater than 1. A PI of less than one would result in a negative return, whereas a PI of one is anticipated to break even. Every company in the project has a PI of more than 1.

IRR is generally the most used profitability metric due to its sophisticated and simple interpretation. However, IRR is difficult to calculate with complex cash flows.

	Initial Investment	Profitability Index	Rank	NPV	IRR	Payback Period	ARR
Project A	-525000	£1.49	1	£255,896.50	21%	11	34%
Project B	-375000	£1.22	4	£84,223.18	11%	10.96	29%
Project C	-448000	£1.15	5	£66,194.85	10%	7.88	26%
Project D	-625000	£1.40	2	£249,737.63	17%	5.12	32%
Project E	-254000	£1.35	3	£89,722.89	17%	2.58	31%

Projects A and D have the highest NPV, IRR, PI, and ARR, according to the calculation. Project D would come in second with a higher return. The business has the option to invest in Project D as well as Project A.

MODEL OVERVIEW

The two models utilised in this framework are the Gordon Growth Model and the CAPM. They fulfil an obligation and are essential to investment analysis and decision-making.

A basic idea called CAPM establishes predicted returns depending on risk. It is assumed that increasing demand for high-risk investments will come from investors.

The Capital Asset Pricing Model (CAPM) is utilised to calculate the cost of equity generated by companies and forecast higher levels of returns on investments at a given level of risk. able to weigh the predicted returns of various investment options to determine which is more appealing.

To compute CAPM: -

E(R_{i}) = Rf+ beta*(E(Rm-Rf))

CAPM CALCULATION		
RF RATE (10 Year Government Bond)		4.10%
RM RATE(Average of FTSE historical data)		6.10%
Beta(Vodafone stock beta)		0.98
CAPM		6.06%

The CAPM uses government bond data and the average of historical FTSE data. It might not match the company's prediction or the values that are current or projected.

In this instance, the yield on the 10-year government bond (4.10%) represents the risk-free rate. The difference between the market's predicted return (as measured by the FTSE index) and the risk-free rate, or 6.10% - 4.10% = 2%, is known as the market risk premium. Vodafone's stock has a beta of 0.98, which indicates that it is not as volatile as the market as a whole.

Thus, based on this computation, the CAPM predicts that the investor will receive an average return on investment of 6.06%.

But there are several limitations with this strategy.

The CAPM is predicated on a number of assumptions, including a perfect market, consistent expectations for future cash flows, and the premise that investors will lend money at a risk-free rate without incurring transaction costs.

It considers both systematic and unsystematic risk equally. Since CAPM shows the market index as a total portfolio, there may be errors in the calculation (KENTON, 2023).

Gordon Growth

The Gordon Growth Model, sometimes referred to as the Dividend Growth Model, calculates the stock's intrinsic value using projected dividend payments. It uses the present value of the future dividend to calculate the stock price and makes the assumption that investors are interested in getting dividends.

By examining potential dividend payments, the Gordon model aids in decision-making and is used to determine the stock's intrinsic value. According to this approach, stocks with consistent dividend payments have the potential to rise.

To compute GGM

P0= D1 / (k - g)

Divident growth Model				
Current divident			0.046	
Divident in 1 year			0.04692	
Constant growth rate			0.02	
Annual required rate of return			0.08	
				price
		p0	0.782	78.2
		ke	8%	

This is the dividend per share that the corporation is currently paying out. The current dividend in this instance is 0.046. The projected dividend in a year is 0.04692 using the present pay-out and a constant growth rate. The lowest rate of return required by investors as payment for purchasing company stock. It is sometimes referred to as the company's equity cost. Price is now trading at 72.32, which is below the Gordon Growth Model's defined price of 78.2.The requisite yearly rate of return in this instance is 8%.

GGM does not apply to companies that do not pay dividends because it is based on the assumption that the company will pay dividends at a consistent rate in the future. The methodology excludes other aspects of the business when valuing and only considers dividends. Even in cases where markets are not efficient, it is assumed that prices would accurately reflect all available information (Institute, 2023)

WACC ANALYSES (CAPM)

Capital budgeting decisions and the profitability of investment projects can both be made using the WACC computation. A project is considered profitable if its anticipated return exceeds its weighted average cost of capital (WACC).

Valuation Assumptions		Data Sources
g (World Economic Growth)	3.0%	
Rf	4.1%	10 year GOI bond yield
Rm	6.1%	10 year CAGR % - FTSE 100
Beta	0.98	
CMP	72.32	
Valuation Data		
Total debt (Long term borrowings) (2023)	66,390	
Cash & Cash equivalents (2023)	11,705	
Number of diluted shares (2023)	27775.0	
Tax rate (2023)	4%	
Interest expense rate (2023)	10.50%	
MV of equity	2,008,688	
Total Debt	66,390	
Total capital	2,075,078	
WACC		
We	96.8%	
Wd	3.2%	
Ke = *Rf + Beta *(Rm-Rf)*	6.1%	
Kd = *Interest (1-t)*	10.1%	
WACC = *(We*ke)+(Wd*kd)*	6.2%	

WACC indicates the overall riskiness of the business and assists in determining the ideal capital structure by balancing the costs of debt and equity. This is influenced by various aspects such as industry dynamics, competitive environment, and financial leverage (CFI, 2023).

This CAPM model WACC estimates an 6.06% discount rate.
A WACC, of 6.06% means that a company's total cost of funding—which takes into account both stock and debt—is 6.06%. It's an essential statistic for capital budgeting decisions and for assessing the profitability of investment projects.

Gordon Growth	
We	96.80%
wd	3.20%
ke	8%
kd	10.10%

(Sources: - Annual Report of Vodafone)

This GGM model WACC estimates an 8.08% discount rate.
A WACC, of 8.08%% means that a company's total cost of funding—which takes into account both stock and debt—is 6.06%. It's an essential statistic for capital budgeting decisions and for assessing the profitability of investment projects.

Additionally, WACC has several limitations because it uses previous data to compute utilising two models that do not precisely reflect future profiles. Additionally, it ignores flaws in the market and behavioural biases that could influence investment choices.

CAPITAL BUDGETING ANALYSES

The process of assessing and choosing long-term investment projects is known as capital budgeting. It is a crucial tool for companies to use when deciding how best to distribute their resources. There exist multiple strategies for capital budgeting, each with unique advantages and disadvantages.

The capital budgeting approaches include ARR, PI, NPV, IRR, and Payback Period. A more sophisticated method of capital budgeting is Monte Carlo simulation, which models the uncertainty of future cash flows using probability distributions. For initiatives with a high level of risk, this can be helpful.

When employing capital budgeting approaches, it is crucial to take into account many elements, including the capital cost, project risk, cash flow timing, and the strategic goals of the company. Early cash flow initiatives are typically more appealing than those with late cash flows (KENTON, 2023).

	Initial Investment	PI (CAPM)	Rank	NPV(CAPM)	IRR	PP(CAPM)	NPV(GGM)	PI(GGM)	PP(GGM)	ARR
Project A	-525000	1.50	1	£263,287.10	21%	4.42	£179,340.56	1.34	3.47	34%
Project B	-375000	1.24	4	£88,878.98	11%	4.38	£36,432.49	1.10	4.62	29%
Project C	-448000	1.16	5	£70,393.07	10%	3.82	£22,807.72	1.05	4.64	26%
Project D	-625000	1.41	2	£258,457.80	17%	3.82	£159,831.13	1.26	3.99	32%
Project E	-254000	1.37	3	£92,899.28	17%	3.67	£56,874.16	1.22	3.87	31%

A discount rate of 6.06% was produced by WACC with CAPM; however, a higher rate of 8.08% was provided by the Gordon Growth Model. All of the investment options had positive NPV and profitability indices prior to being analysed using capital budgeting methodologies, which is a positive indication to invest in any of the aforementioned portfolios. Project A has the highest net present value (NPV) of any investment choice in both models, with NPVs of 263,287 (CAPM) and 179340 (GGM), respectively.

With an NPV of £258,457.80 in the CAPM and £159,831.13 in the GGM model, Project D has the second-highest net present value (NPV). Its projected internal rate of return (IRR) is 17%, which is 4% lower than that of Project A.

With an NPV of £92,899.28 in the CAPM and £56,874.16 in the GGM model, Project E is regarded as the greatest investment choice after Projects A & D. Project D and E share the same IRR rate.

Project E has the lowest NPV of all the investment options at £70,393.07, while Project B has the highest NPV of £88,878.98 (CAPM) and the highest IRR of 11%. This demonstrates that Project A is the greatest choice for investment based on capital budget rationing approaches, or that Project D and A are investments that a corporation with excess income can make, respectively.

CONCLUSION

In summary, capital budgeting has advantages and disadvantages of its own when compared to other strategies. However, the success of a given project will also rely on its unique features and the company's level of risk tolerance. While the GGM and CAPM models are dependable, the results are limited. Given that the GGM approach has a larger WACC than the CAPM method, the business should concentrate on the method with a lower WACC. The Vodafone FC has two options: either invest in Project A, which has a larger net present value, or diversify their holdings in Projects A and D.

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